

Magnetic Bloch states of 2D Dirac fermions in a mass superlattice

Aman Anand

aman.anand@citystgeorges.ac.uk

The University of
business, practice
and the professions.

citystgeorges.ac.uk

Special problems in Condensed Matter Physics
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Work in collaboration with: **A. De Martino, R. Egger**

- ❖ Why Dirac fermions?
- ❖ What happens to Dirac fermions in different mass profiles under a magnetic field?
- ❖ Are there unique signatures in transport?

Crystals & Translational Symmetry

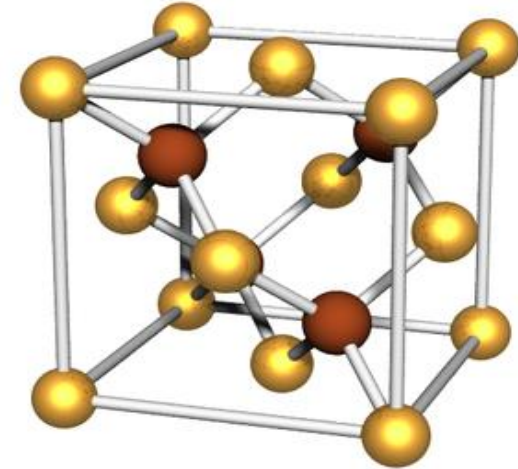
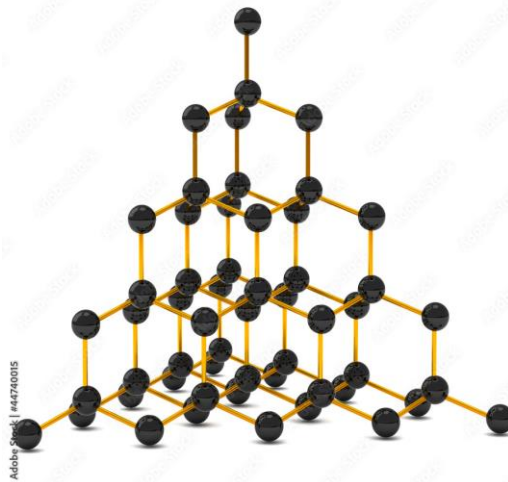
Bloch's theorem:

$$\diamond [\hat{H}, \hat{T}_R] = 0$$

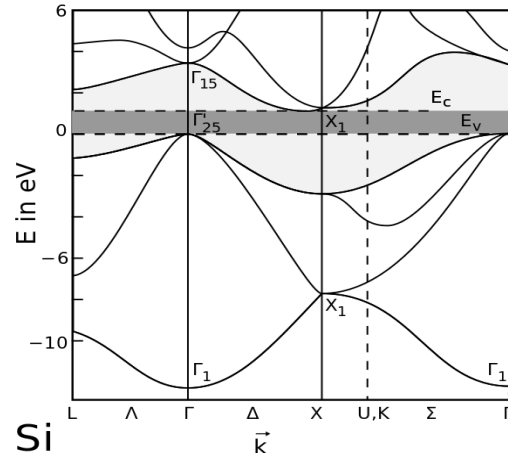
$$\diamond \hat{T}_a f(\mathbf{r}) = f(\mathbf{r} + \mathbf{a})$$

$$\diamond \hat{T}_R \psi(\mathbf{r}) = e^{i\mathbf{k} \cdot \mathbf{R}} \psi(\mathbf{r})$$

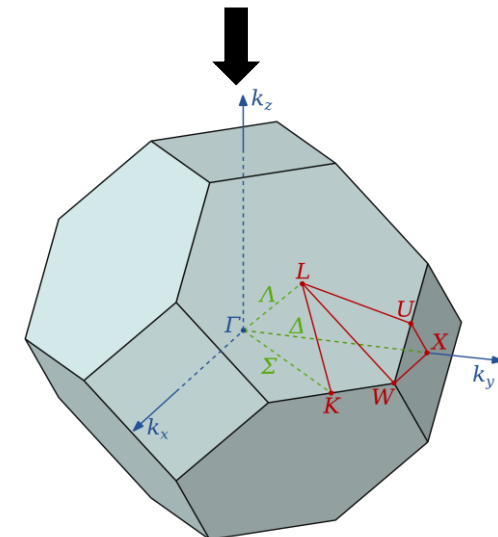
$$\diamond \psi(\mathbf{r}) = e^{i\mathbf{k} \cdot \mathbf{r}} u_{\mathbf{k}}(\mathbf{r})$$



<http://gisaxs.com/index.php/Lattice:Diamond>

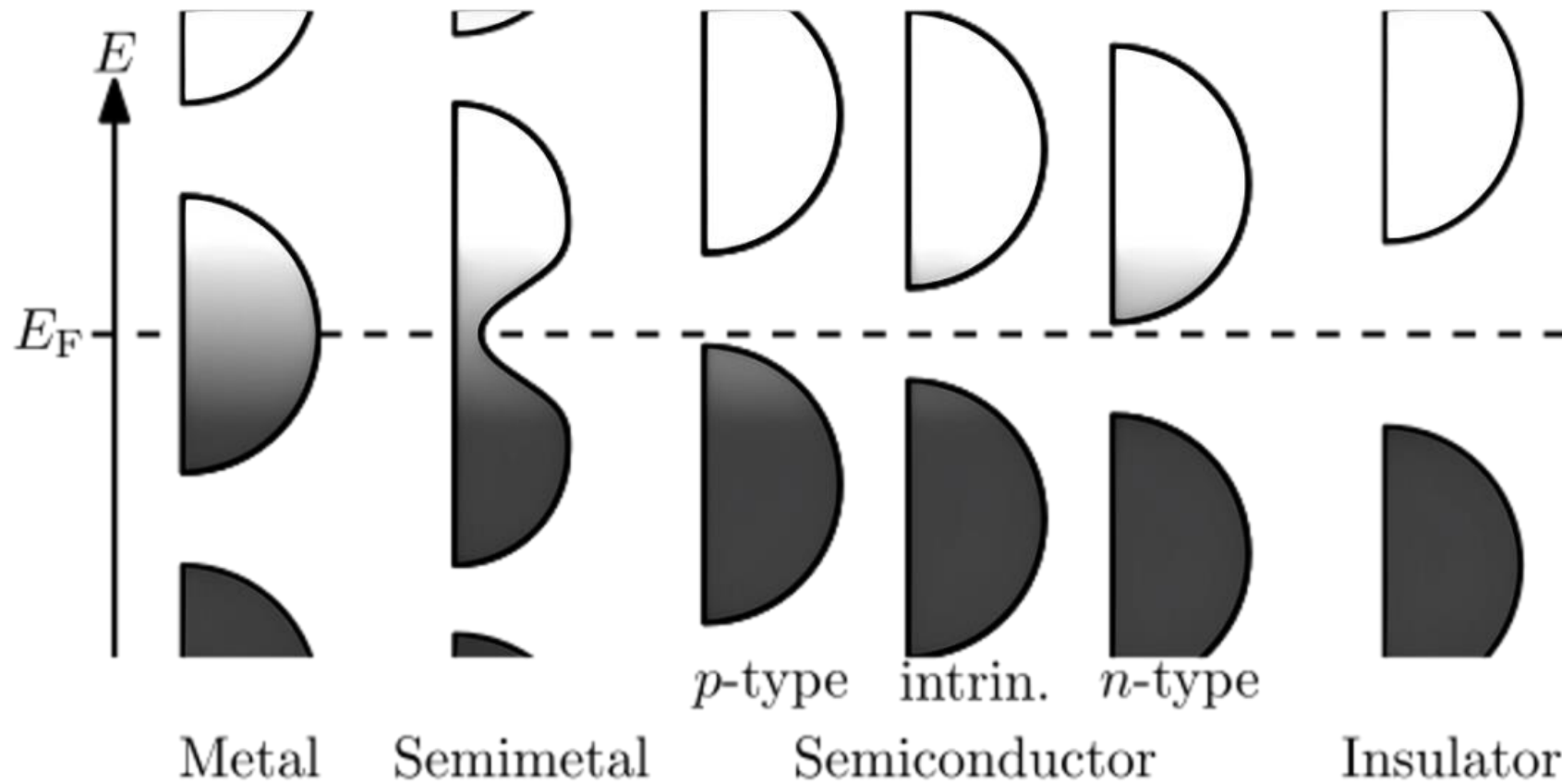


https://commons.wikimedia.org/wiki/File:Band_structure_Si_schematic.svg



https://fr.m.wikipedia.org/wiki/Fichier:Brillouin_Zone_%281st_FCC%29.svg

Bandstructure



https://en.wikipedia.org/wiki/Semimetal#/media/File:Band_filling_diagram.svg

- ❖ The Dirac equation in $d + 1$ dimensions

$$(i \gamma^\mu \partial_\mu - m) \psi = 0$$

where $\mu = 0, 1, \dots, d$

- ❖ For $d = 1$,

$$i \partial_t \psi = (-i \gamma^0 \gamma^1 \partial_x + m \gamma^0) \psi$$

where $\gamma^0 = \sigma_z$, $\gamma^1 = i \sigma_y$, and $\gamma^5 = \gamma^0 \gamma^1$

- ❖ $\{\gamma^\mu, \gamma^\nu\} = 0, \quad \mu \neq \nu$

- ❖ $(\gamma^0)^2 = -(\gamma^i)^2 = \text{Id}$

- ❖ If $d = 2k + 1$ or $2k + 2$, then γ is a matrix of size $(2^{k+1} \times 2^{k+1})$

- ❖ For $d = 2$,

with $\gamma^0 = \sigma_z$, $\gamma^1 = i \sigma_y$, and $\gamma^2 = -i \sigma_x$

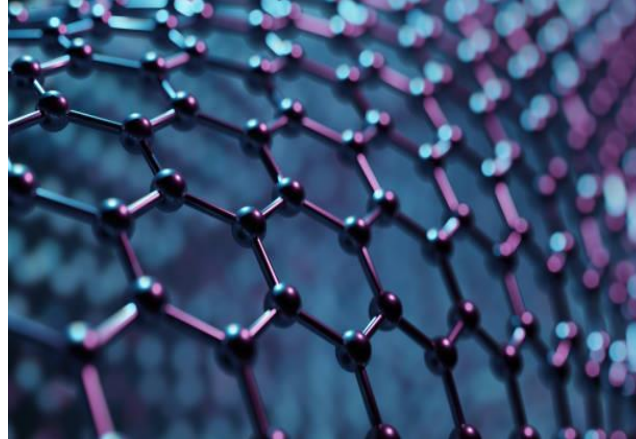
$$i \partial_t \psi = (-i \sigma_x \partial_x - i \sigma_y \partial_y + m \sigma_z) \psi$$

[A. Vishwanath](#), et al.: Rev. Mod. Phys 90, 015001 (2018)

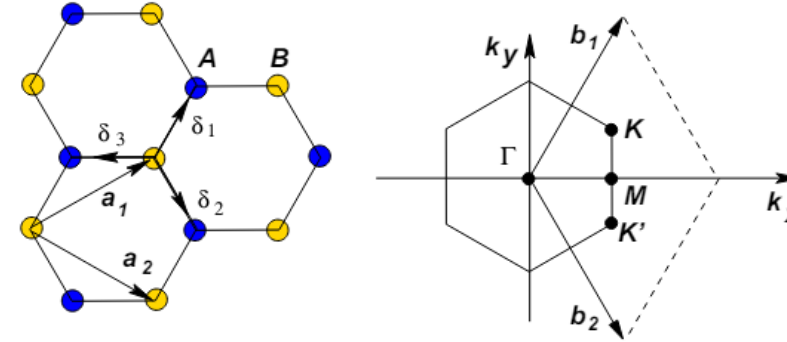
[P.A.M.Dirac](#): Proceedings of Royal Society of London, Vol. 117, No. 778, pp. 610-624 (1928)

Graphene

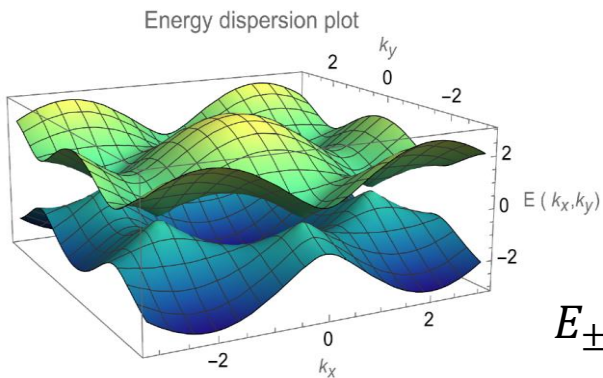
- ❖ First 2D material
- ❖ Energy sector, Membranes for desalination, gas sensors, microwave adsorption, biology



iStock Credit: vchal

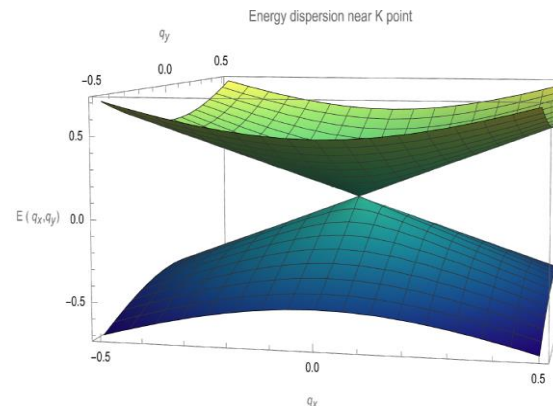


$$H = -t \sum_{\langle i,j \rangle} (a_i^\dagger b_j + h.c.) - t' \sum_{\langle\langle i,j \rangle\rangle} (a_i^\dagger a_j + b_i^\dagger b_j + h.c.)$$



$$E_{\pm}(\mathbf{k}) = \pm t \sqrt{3 + f(\mathbf{k})} - t' f(\mathbf{k})$$

$$f(\mathbf{k}) = 4 \cos\left(\frac{\sqrt{3}}{2} k_y a\right) \cos\left(\frac{3}{2} k_x a\right) + 2 \cos(\sqrt{3} k_y a)$$



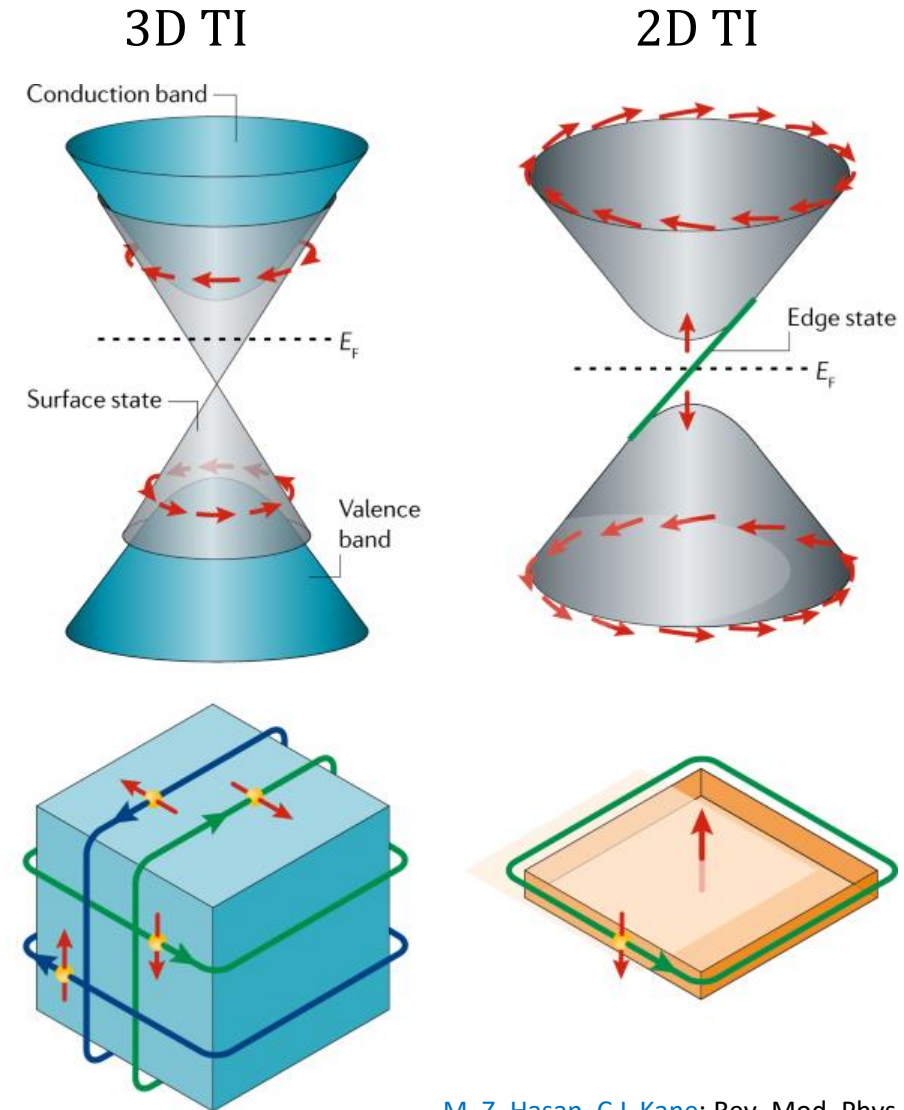
$$E_{\pm}(\mathbf{q}) = \pm v_F |\mathbf{q}| = \pm v_F \sqrt{q_x^2 + q_y^2}$$

$$H_{eff.} = \hat{p}_x \sigma_x + \hat{p}_y \sigma_y$$

[Castro Neto](#), et al.: Rev. Mod. Phys. 81, 109 (2009)

Topological Insulator Edge

- ❖ Insulating bulk; conducting edges
- ❖ 3D TI : Bi_2Se_3 , Sb_2Te_3
- ❖ 2D TI : HgTe/CdTe quantum wells
- ❖ **Surface states** : Spin momentum locking; robust against local symmetry preserving perturbations
- ❖ $H_{\text{surf.}} = v_F(k_y\sigma_x - k_x\sigma_y)$
with $E(\mathbf{k}) = \pm v_F|\mathbf{k}|$



[M. Z. Hasan, C.L.Kane](#): Rev. Mod. Phys. 82.3045 (2010)
[Y. Tokura, K. Yasuda, A. Tsukazaki](#): Nat Rev Phys 1,126–143 (2019)

Getting mass by symmetry breaking

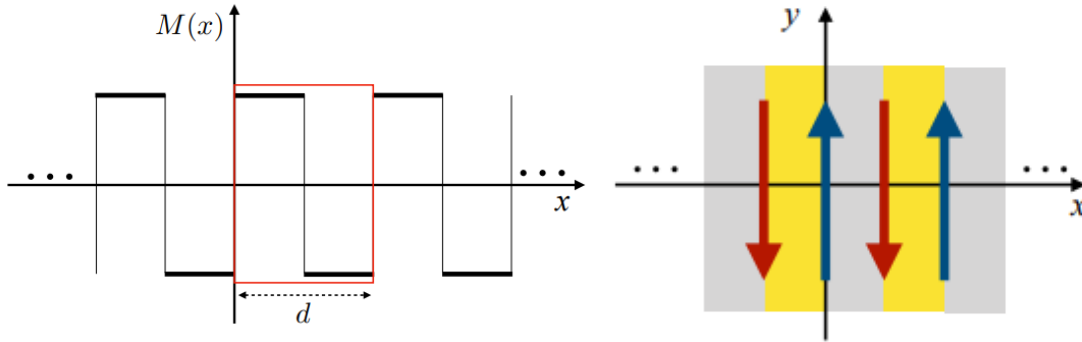
- ❖ For **Topological Insulator (TI)** surface, we can break the **Time Reversal** symmetry by **doping with magnetic impurities** or **depositing ferromagnetic layer** to get a mass term like

$$H_{surf. TI} = v_F(k_y\sigma_x - k_x\sigma_y) + m\sigma_z$$

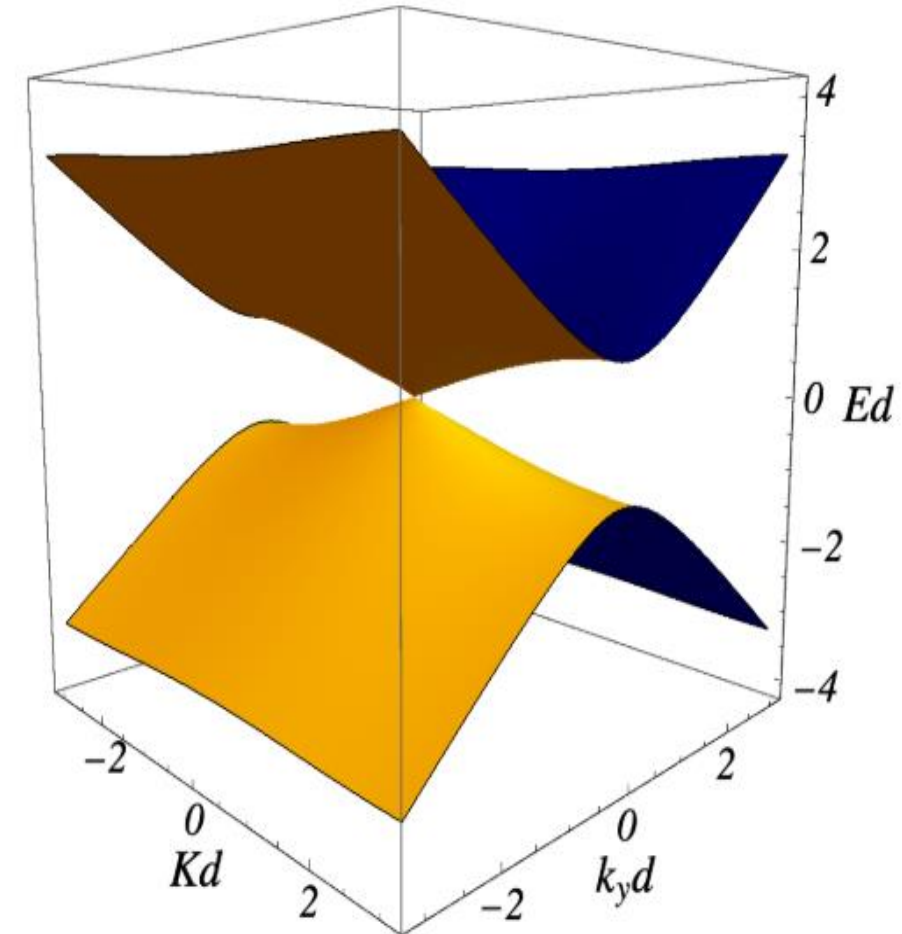
- ❖ For **Graphene**, we can break the **sublattice** symmetry with **strain** or **substrates**, to get a mass term like

$$H_{graphene} = v_F(k_x\sigma_x + k_y\sigma_y) + m\sigma_z$$

2D-Dirac fermions in mass superlattice



- ❖ $H = -i\partial_x\sigma_x - i\partial_y\sigma_y + m(x)\sigma_z$
- ❖ **Mass kink** $\Rightarrow$ Localized in one direction + **Chiral** propagation in other: $E(k_y) \propto k_y$
- ❖ **Periodic in x, Translational invariance in y**; good quantum numbers: quasi-momentum K along x and k_y in y.
- ❖ **Highly anisotropic Dirac cone** ($v_x \ll v_y = v_F$)



A. De Martino, L. Dell'Anna, L. Handt, A. Misrocchi, and R. Egger: Phys. Rev. B **107**, 115420

Dirac fermions with mass in magnetic field

- ❖ In Landau gauge

$$H(k_y) = -i\partial_x\sigma_x + (k_y + eBx)\sigma_y + m(x)\sigma_z$$

- ❖ For a constant mass M

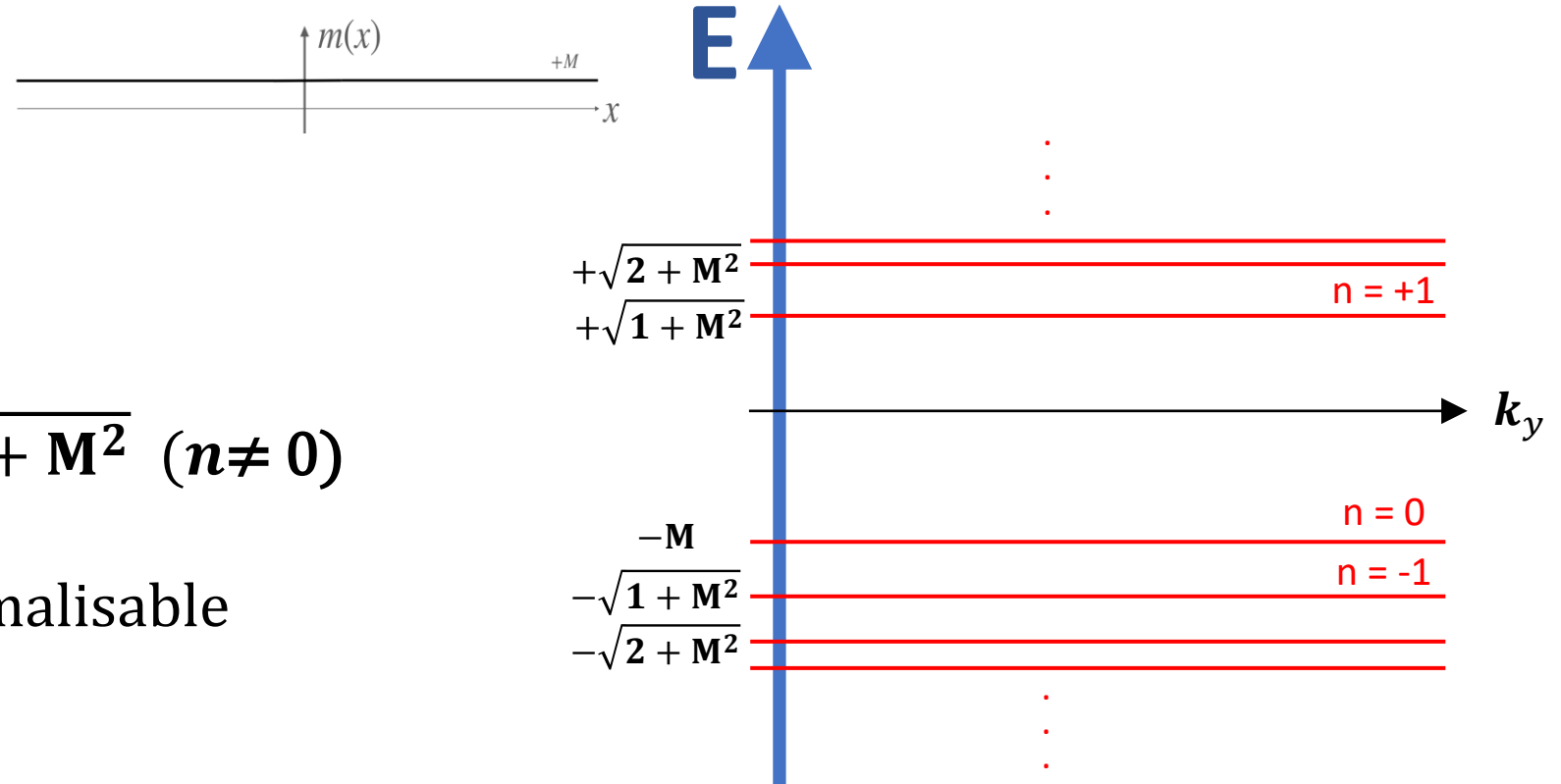
$$\psi(x) = W_M(x) \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} (E + M)D_{p-1}(-q) & (E + M)D_{p-1}(q) \\ -iD_p(-q) & iD_p(q) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}, \quad \begin{aligned} p &= E^2 - M^2 \\ q &= \sqrt{2}(k_y + x) \end{aligned}$$

where $D_p(x)$ are the parabolic cylinder functions

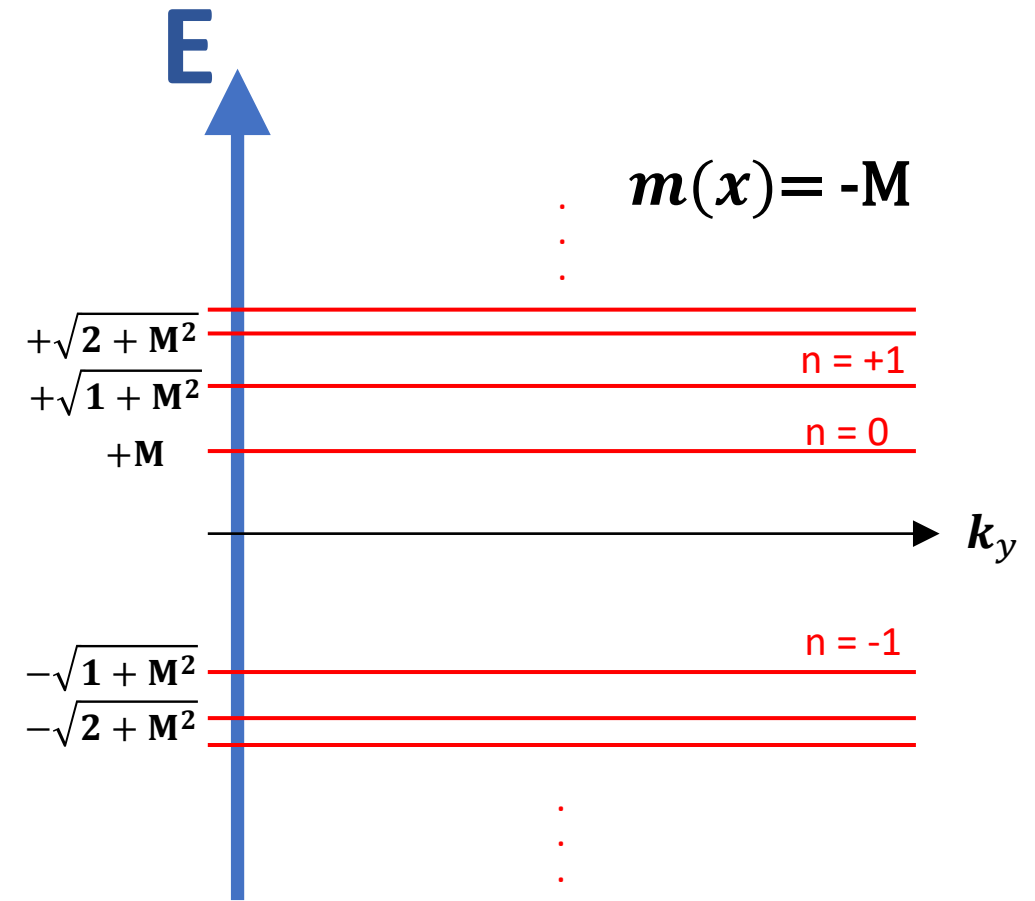
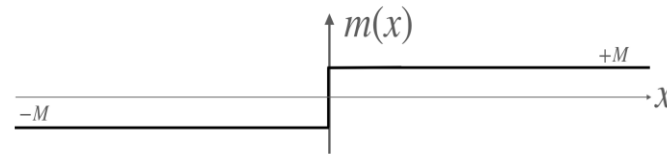
- ❖ **Mass, energy** in units of ε_c and **length** in units of l_B , with $l_B = \sqrt{\frac{c}{eB}}$ and $\varepsilon_c = \frac{\sqrt{2}v_F}{l_B}$.

Dirac-Landau levels with uniform mass

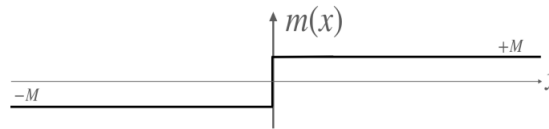
- ❖ Uniform mass M
- ❖ $E_0 = -M$
 $E_n = \text{sgn}(n)\sqrt{|n| + M^2} \quad (n \neq 0)$
- ❖ $E = M$ is non-normalisable



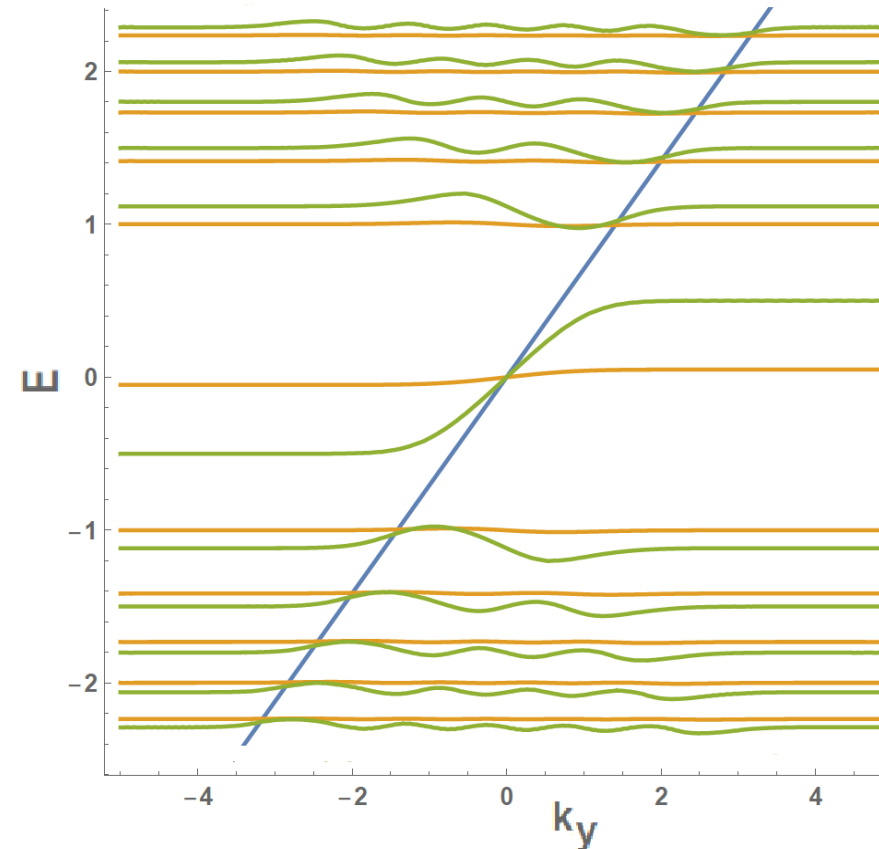
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Mass kink contd.



- ❖ Velocity of Jackiw-Rebbi modes can be tuned.
- ❖ Max. velocity of propagation in **blue** for very large mass.
- ❖ Small mass resembles **flat** Landau levels in graphene.

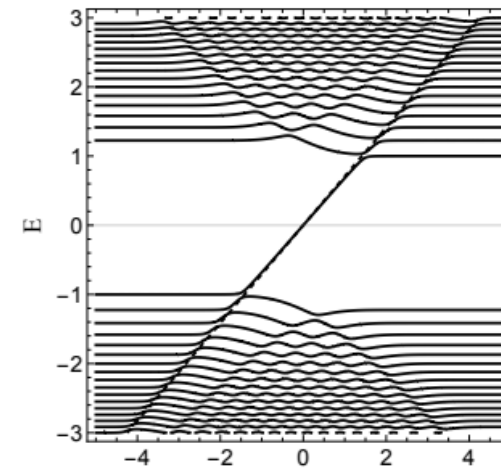


$M = 0.05$

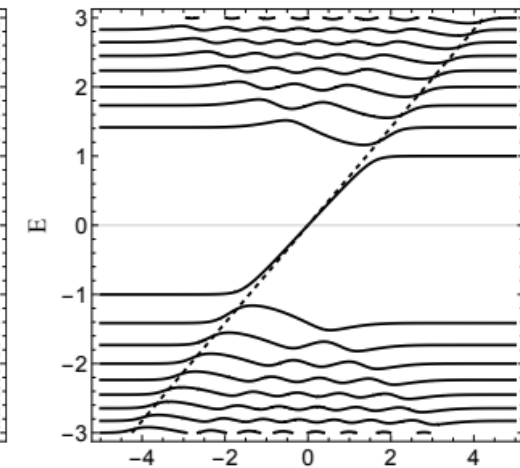
$M = 0.5$

Mass kink contd.

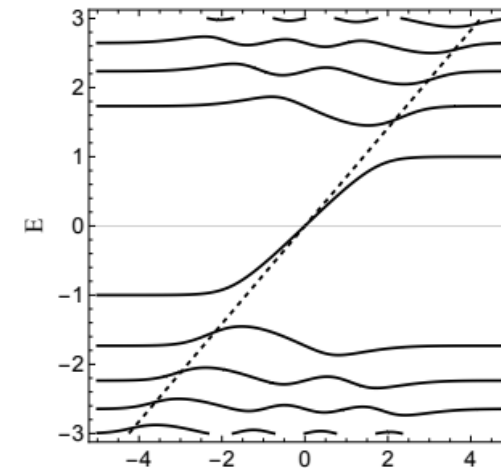
- ❖ Velocity of Jackiw-Rebbi modes can be tuned by changing the magnetic field.
- ❖ Max. velocity of propagation in dotted for $B = 0$.
- ❖ Stronger magnetic field decreases velocity of Jackiw-Rebbi modes.



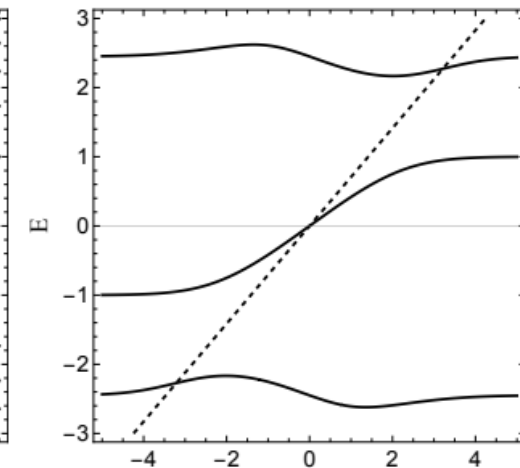
$B = 0.5T$



$B = 1.0T$



$B = 2.0T$



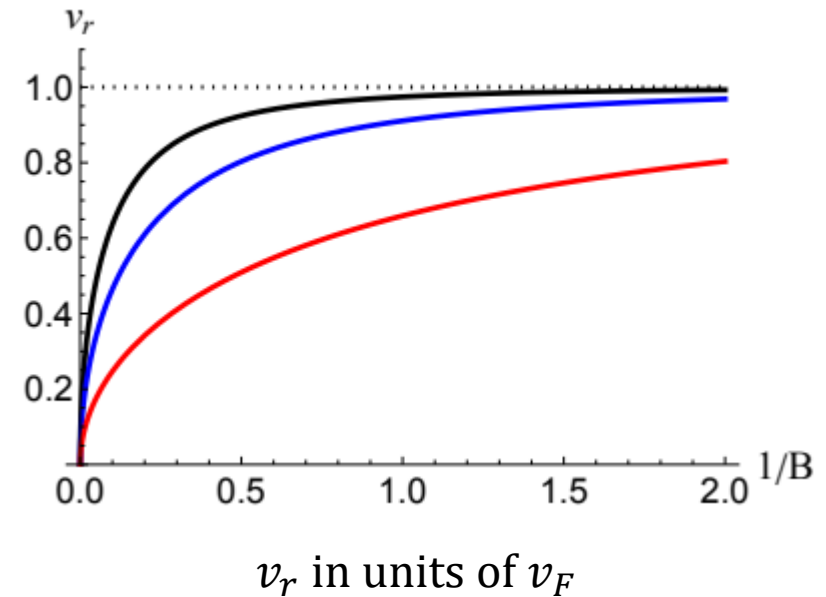
$B = 5.0T$

Renormalized velocity of Chiral modes

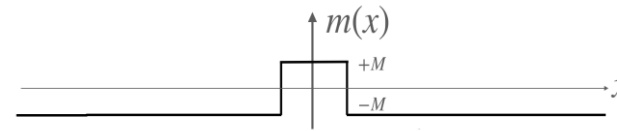
❖ $v_r(\zeta) = 2^{\frac{3}{2}-\zeta^2} \sqrt{\pi} \zeta \left(\frac{\Gamma(1+\zeta^2)}{\Gamma^2\left(\frac{1+\zeta^2}{2}\right)} - \frac{\Gamma(1+\zeta^2)}{\Gamma\left(\frac{\zeta^2}{2}\right)\Gamma\left(\frac{2+\zeta^2}{2}\right)} \right)$ with $\zeta = \frac{M}{\varepsilon_c} = \frac{M}{\sqrt{2eB}}$

❖ For $\zeta \ll 1$, $v_r \sim \sqrt{\frac{8}{\pi}} \frac{M}{\varepsilon_c}$ approaches 0

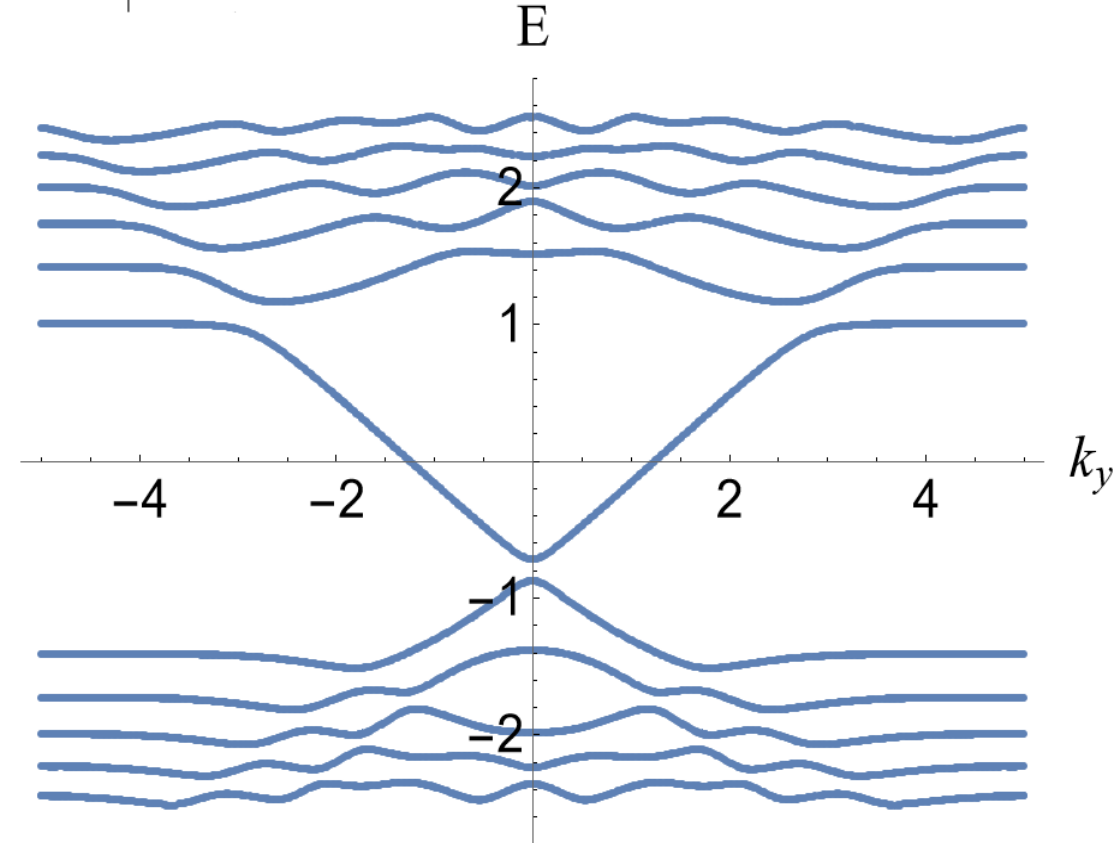
❖ For $\zeta \gg 1$, v_r approaches v_F



Mass barrier

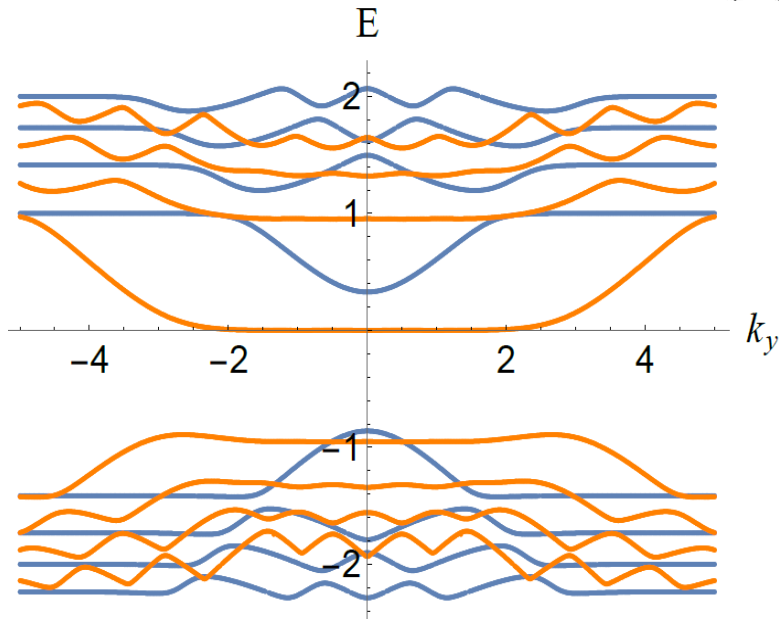
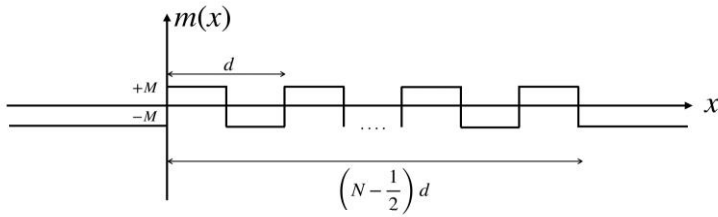


- ❖ Exact solutions using transfer matrix approach.
- ❖ The wavefunction at given E and k_y , localised at $x = -k_y l_B^2$
- ❖ Counterpropagating modes separated only if $d > l_B$

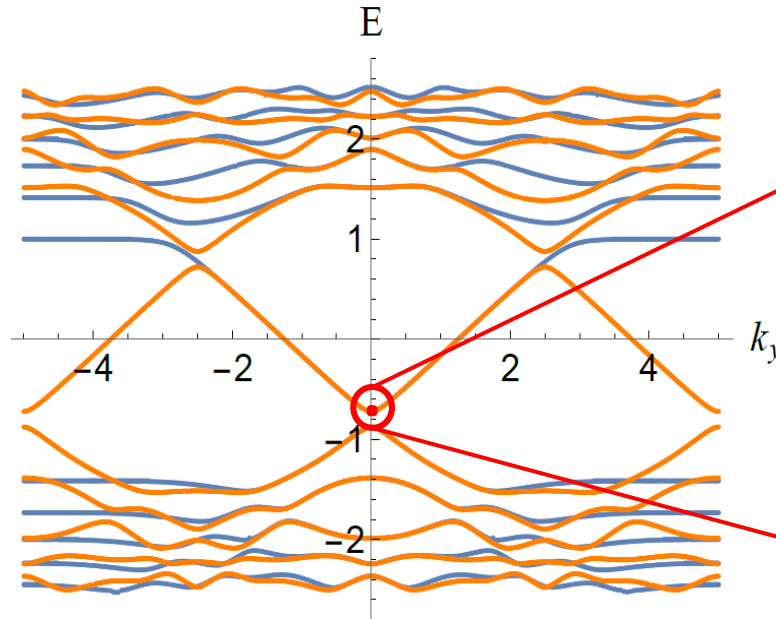


$$M = 1$$
$$d = 5$$

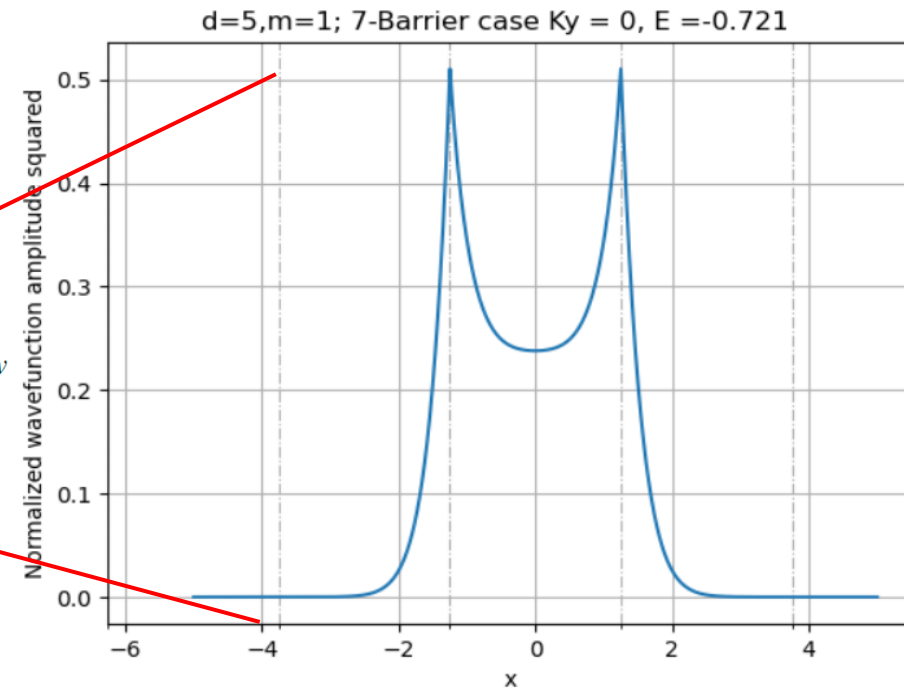
Finite array



$M=1, d=1$



$M=1, d=5$



7-barriers, 1-barrier

Projection on magnetic Bloch states

- ❖ Without fixing the gauge

$$H(k_y) = \underbrace{(-i\partial_x + eA_x)\sigma_x + (-i\partial_y + eA_y)\sigma_y}_{H_0} + \underbrace{m(x)\sigma_z}_{U(r)}$$

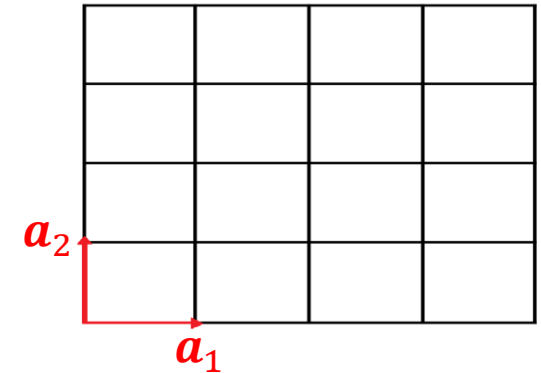
- ❖ Canonical momenta: $\Pi_\mu = -i\partial_\mu + eA_\mu$

- ❖ Guiding centre momenta: $Q_\mu = -i\partial_\mu + eA_\mu + eB\epsilon_{\mu\nu}x_\nu$

- ❖ Ladder operators: $a^\dagger = \frac{\Pi_x + i\Pi_y}{\sqrt{2eB}}$, $a = \frac{\Pi_x - i\Pi_y}{\sqrt{2eB}}$, $b^\dagger = \frac{(\mathbf{a}_1 - i\mathbf{a}_2) \cdot \mathbf{Q}}{\sqrt{2\phi}}$, $b = \frac{(\mathbf{a}_1 + i\mathbf{a}_2) \cdot \mathbf{Q}}{\sqrt{2\phi}}$

- ❖ $H_0 = \varepsilon_c \begin{pmatrix} 0 & a \\ a^\dagger & 0 \end{pmatrix}$, and $[a, a^\dagger] = [b, b^\dagger] = 1$, all a and b ladder operators

commute among each other.

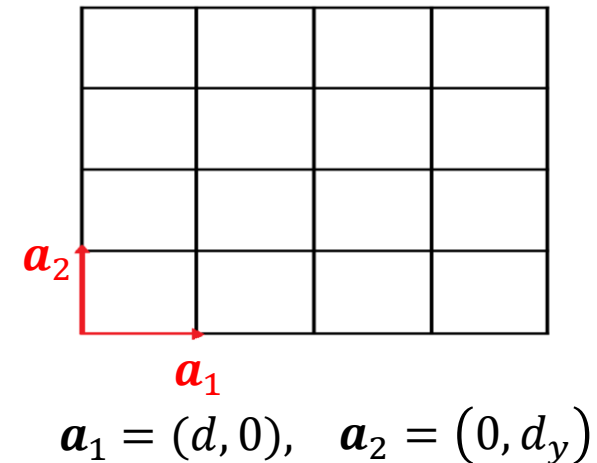


$$\mathbf{a}_1 = (d, 0), \quad \mathbf{a}_2 = (0, d_y)$$

$$\phi = \frac{e}{\hbar} \Phi = \frac{eB |\mathbf{a}_1 \times \mathbf{a}_2|}{\hbar}$$

Projection on magnetic Bloch states contd.

- ❖ $H_0|\psi_n\rangle = E_n|\psi_n\rangle$, with $E_n = \text{sgn}(n) \sqrt{2|n|eB}$
- ❖ $|\psi_n\rangle = \mathcal{N}_n \binom{\text{sgn}(n)|n|-1,0\rangle}{|n|,0\rangle}$, with $\mathcal{N}_n = \frac{1}{\sqrt{2-\delta_{n,0}}}$, and $|n,m\rangle = \frac{a^\dagger}{\sqrt{n!}} \frac{b^\dagger}{\sqrt{m!}} |0,0\rangle$ and the state $|0,0\rangle$ is defined by $a|0,0\rangle = b|0,0\rangle = 0$.
- ❖ **Magnetic Translation operators:** $T_{a_i} = \exp(i\mathbf{a}_i \cdot \mathbf{Q})$
- ❖ Translation operators commute with both the kinetic part and any potential part with the lattice periodicity.
- ❖ $T_{a_1} T_{a_2} = e^{-i\phi} T_{a_2} T_{a_1}$, where $\phi = \frac{e}{\hbar} \Phi = \frac{eB |\mathbf{a}_1 \times \mathbf{a}_2|}{\hbar}$
- ❖ Translational operators to commute ($\phi = 2\pi$) $\Rightarrow d_y = \frac{2\pi}{eBd}$



Projection on magnetic Bloch states contd.

❖ Magnetic Bloch states:

$$|\mathbf{k}, n\rangle = \frac{1}{\sqrt{N(\mathbf{k})}} \sum_{\mathbf{R}} e^{-i\mathbf{k} \cdot \mathbf{R}} T_{\mathbf{a}_1}^{\mathbf{R} \cdot \mathbf{b}_1} T_{\mathbf{a}_2}^{\mathbf{R} \cdot \mathbf{b}_2} |\psi_n\rangle$$

$\mathbf{R} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2$ is a general lattice vector,

$\mathbf{b}$ are reciprocal lattice vectors.

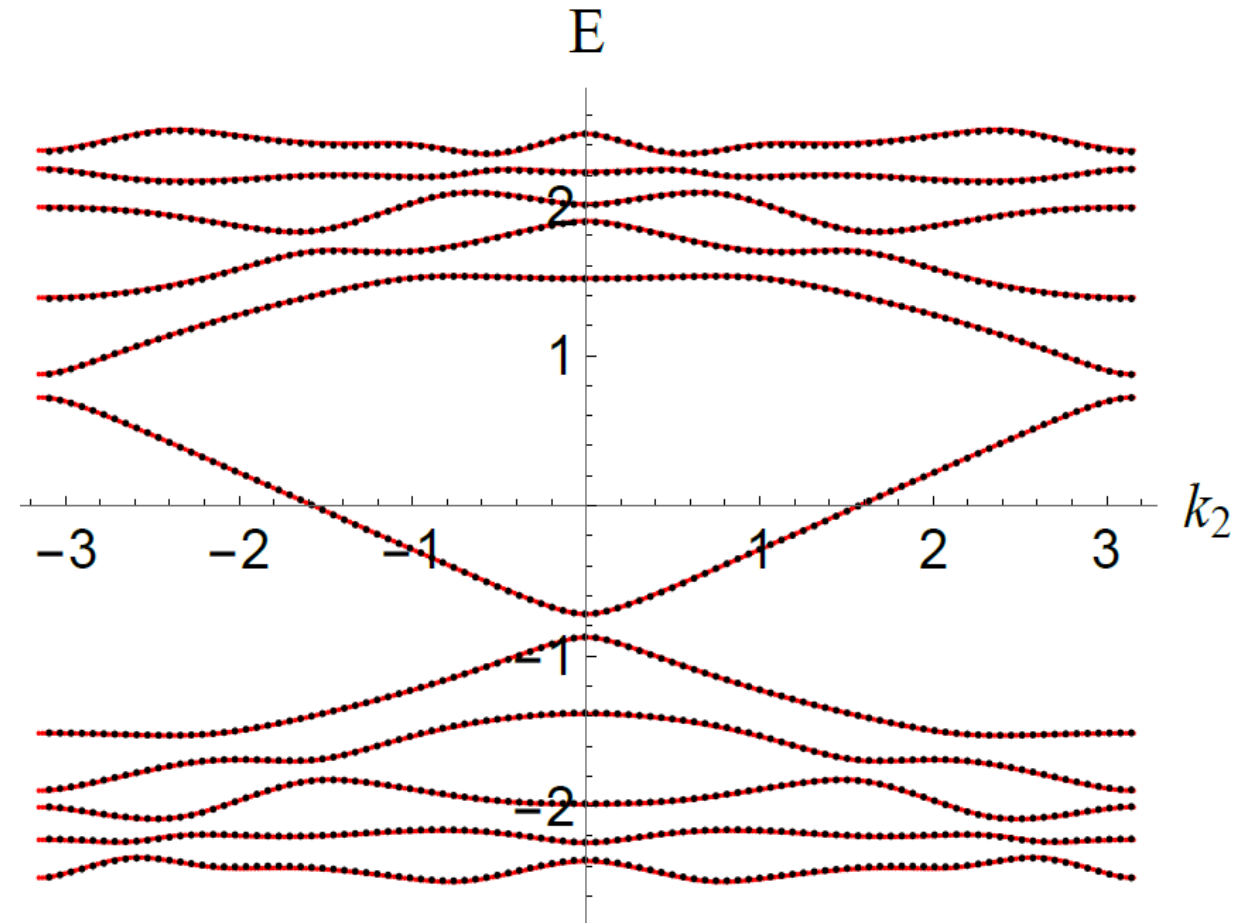
❖ Simultaneous eigenvectors of Hamiltonian and translational operators

$$T_{\mathbf{a}_i} |\mathbf{k}, n\rangle = e^{i\mathbf{k} \cdot \mathbf{a}_i} |\mathbf{k}, n\rangle$$

❖ Orthonormality condition: $\langle \mathbf{k}, m | \mathbf{k}', n \rangle = (2\pi)^2 \delta_{mn} \delta(\mathbf{k} - \mathbf{k}')$

Projection on magnetic Bloch states contd.

- ❖ $\langle \mathbf{k}, m | H | \mathbf{k}, n \rangle = H_{mn}$
- ❖ Degeneracy along $k_1 = k_x d$ direction.
- ❖ Perfect agreement with finite array case.
- ❖ Our procedure works on any 1d mass superlattice.
- ❖ $|\Psi_1(\mathbf{k})\rangle = \sum_n c_n^l(\mathbf{k}) |\mathbf{k}, n\rangle$

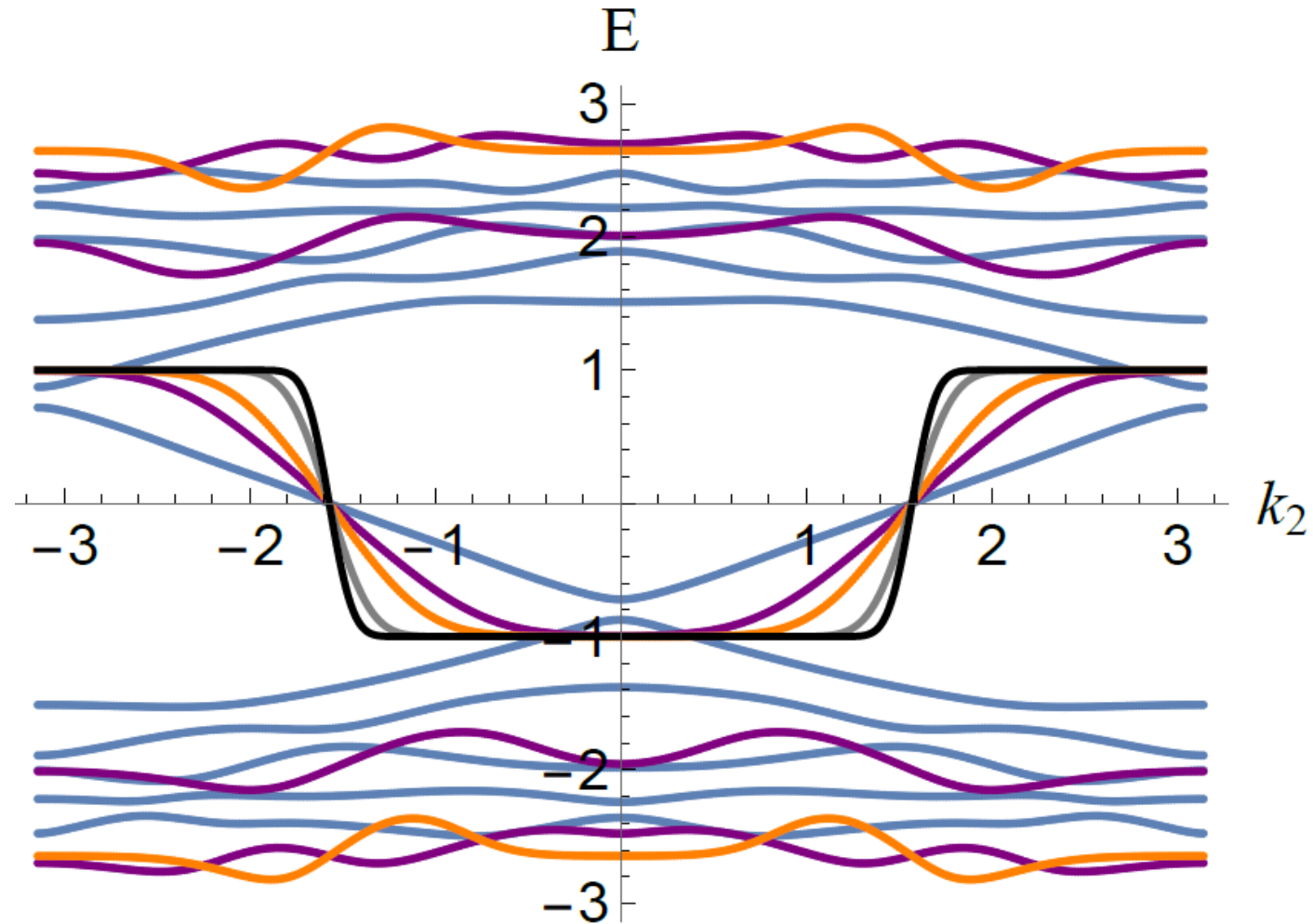


$M = 1$
 $d = 5$

($k_2 = k_y d_y$)

Effect of magnetic field on bandstructure

- ❖ $M=1, d=5$
- ❖ Renormalization of Jackiw-Rebbi modes.
- ❖ Large magnetic field makes band flat in both directions.



$B = 1\text{T}$

$B = 3\text{T}$

$B = 9\text{T}$

$B = 27\text{T}$

$B = 81\text{T}$

- ❖ Hall conductivity
- ❖ Weiss oscillations

Hall conductivity

❖ Kubo conductivity formula:

$$\sigma_{xy} = e^2 \hbar \sum_{l, l' \neq l} \int \frac{d\mathbf{k}}{(2\pi)^2} [f(\varepsilon_l(\mathbf{k})) - f(\varepsilon_{l'}(\mathbf{k}))] \Omega_{l, l'}^{xy}(\mathbf{k})$$

Fermi-Dirac distribution

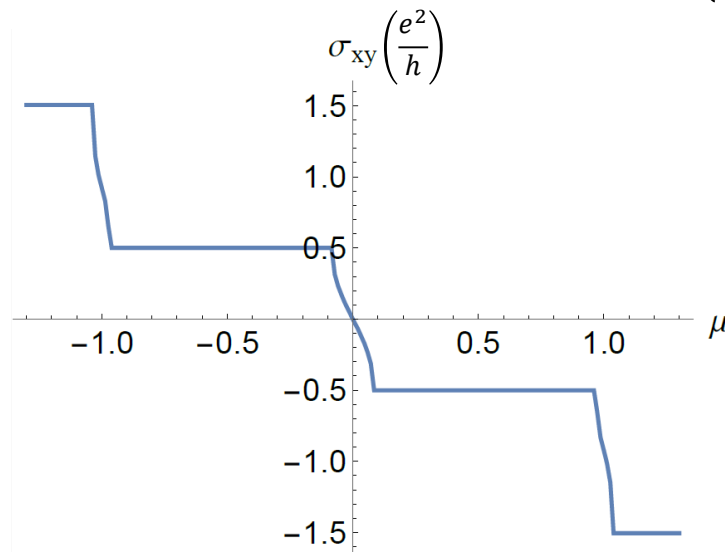
Berry curvature

Band-index

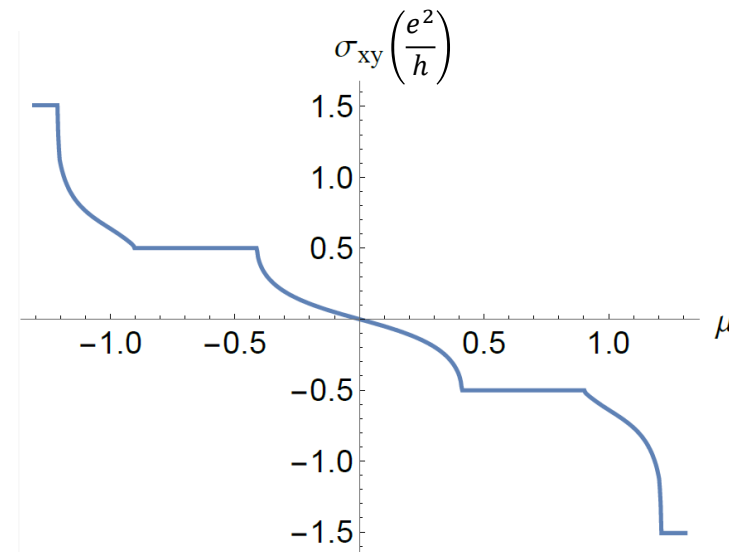
$$\text{where } \Omega_{l, l'}^{xy}(\mathbf{k}) = \text{Im} \frac{\langle \Psi_l(\mathbf{k}) | v_x | \Psi_{l'}(\mathbf{k}) \rangle \langle \Psi_{l'}(\mathbf{k}) | v_y | \Psi_l(\mathbf{k}) \rangle}{(\varepsilon_l(\mathbf{k}) - \varepsilon_{l'}(\mathbf{k}))^2}$$

$$|\Psi_l(\mathbf{k})\rangle = \sum_n c_n^l(\mathbf{k}) |k, n\rangle$$

M=0.1
d = 5



M=0.5
d = 5



[N.P. Ong](#), et al.: Rev. Mod. Phys. 82.1539 (2010)

- ❖ **Weiss oscillations** are oscillations in magnetoresistance, first discovered by Dieter Weiss in 1989.
- ❖ **2DEG** with artificially created periodic potential using interfering laser beams. $V(x) \propto \cos\left(\frac{2\pi x}{d}\right)$
- ❖ Oscillations in magnetoresistance with period $\propto \frac{1}{B}$ (like SdH)
- ❖ **Commensurability condition:** $2R_c = \left(n + \frac{1}{2}\right) d$, with $R_c = \frac{m^* v_F}{eB}$, where m^* is the electron effective mass.

Weiss oscillations in graphene

$$\diamond \sigma_{yy} = \frac{\beta e^2}{L_x L_y} \sum_{\zeta} f(E_{\zeta}) [1 - f(E_{\zeta})] \tau(E_{\zeta}) \left(v_y^{\zeta} \right)^2 \quad \text{with } v_y^{n,k_y} = \frac{1}{\hbar} \frac{\partial \Delta E_{n,k_y}}{\partial k_y}$$

$$\diamond \Delta E_{n,k_y} = \langle \Psi_{n,k_y} | V(x) | \Psi_{n,k_y} \rangle = \frac{V_0}{2} \cos(k_y d) e^{-u/2} [L_{n-1}(u) + L_n(u)]$$

with $u = \frac{d^2}{2l_B^2}$, $L_n(x)$ are Laguerre polynomials and $\Psi_{n,k_y}(x)$ are Landau level wavefunctions of Dirac equation.

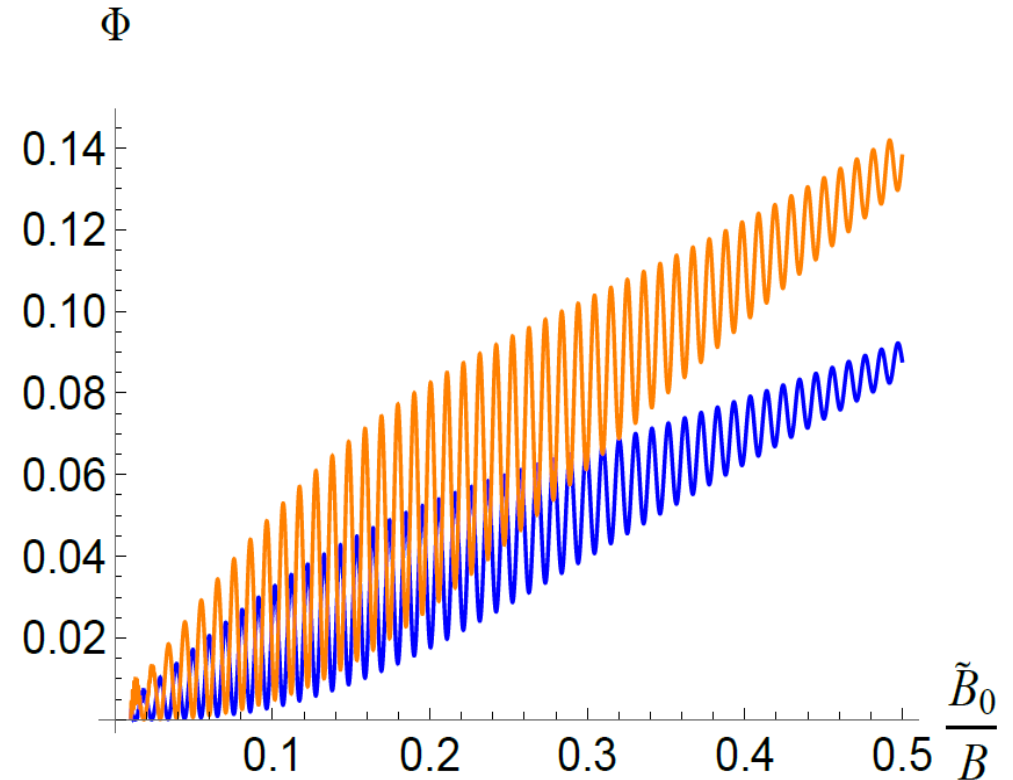
$$\diamond \sigma_{yy} = A \Phi, \quad \text{with } A = \frac{e^2}{h} \frac{M^2 \tau}{\hbar k_B T} \text{ and } \Phi = \frac{1}{4} u e^{-u} \sum_n \frac{g(E_n)}{(1+g(E_n))^2} [L_{n-1}(u) + L_n(u)]^2$$

with $g(E_n) = e^{\beta(E-E_F)}$

$$\diamond \text{Commensurability condition: } 2R_c = \left(n + \frac{1}{2} \right) d, \text{ with } R_c = \frac{\hbar k_F}{eB}, \text{ where } m^* \text{ is the electron effective mass.}$$

Weiss-like oscillations with mass term

- ❖ Mass term like $M \cos\left(\frac{2\pi x}{d}\right) \sigma_z$
- ❖ $\sigma_{yy} = A \Phi$, with $A = \frac{e^2}{h} \frac{M^2 \tau}{\hbar k_B T}$
- ❖ $\Phi = \frac{1}{4} u e^{-u} \sum_n \frac{g(E_n)}{(1+g(E_n))^2} [L_{n-1}(u) - L_n(u)]^2$
- ❖ $\frac{\pi}{2}$ - phase shift because of $m\sigma_z$ term.
- ❖ Decrease in amplitude of oscillations.



$$T = 6\text{K}, \mu = 90.4\text{meV}, \\ d = 358\text{nm}, \tilde{B}_0 = \frac{\hbar}{ed^2}$$

- ❖ Magnetic field **renormalizes** Jackiw-Rebbi propagation velocity.
- ❖ Various **regimes** in spectrum depending on **ratio** between **magnetic length** and the mass **superlattice period**.
- ❖ Observation in **transport** properties like: **Quantum Hall Plateau width** and **Phase shift** in Weiss oscillations.

Thank you!



Dirac cones

+ Mass
→
kink

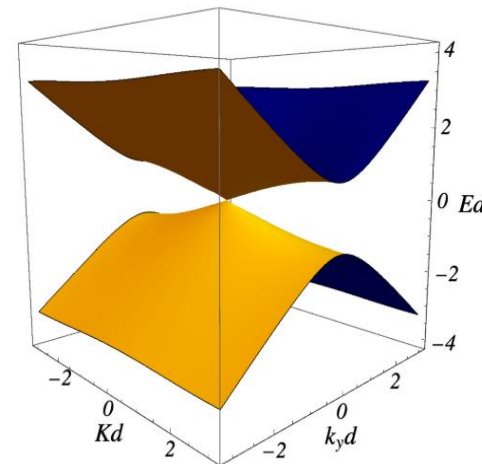
Localized in x

Chiral propagation in y

$$E(k_y) \propto k_y$$

Summary

↓ +Mass superlattice



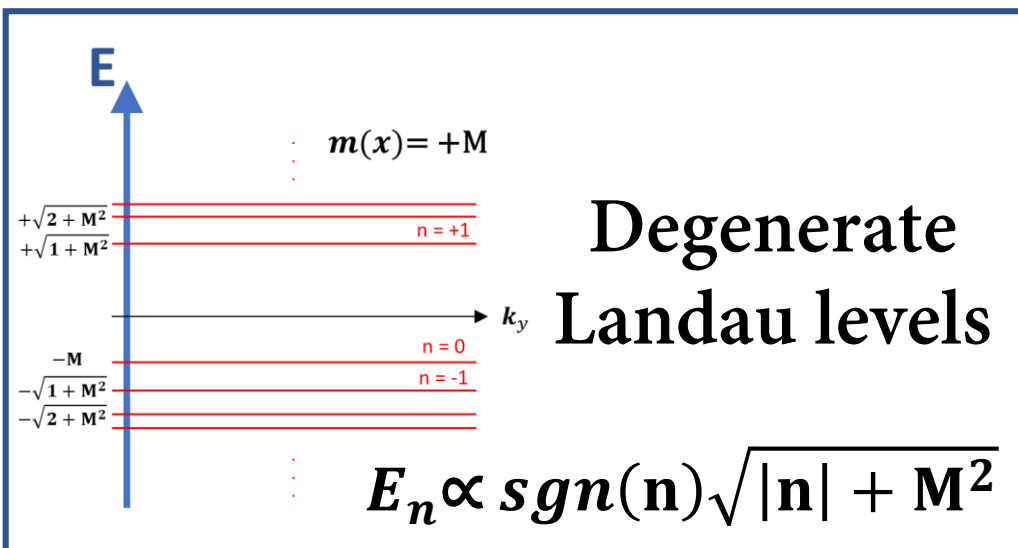
Highly
anisotropic
Dirac cone



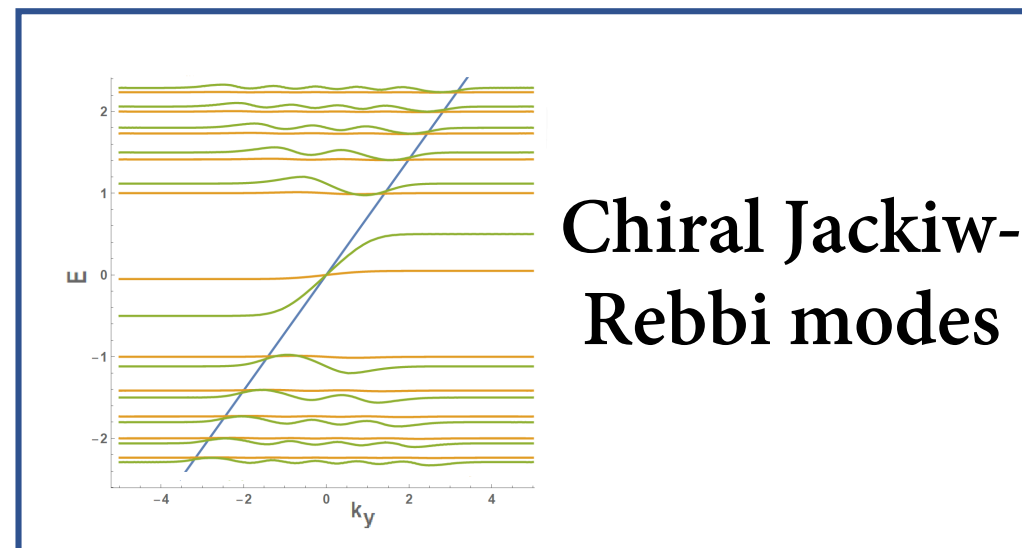
Dirac cones



+Magnetic field
+uniform mass



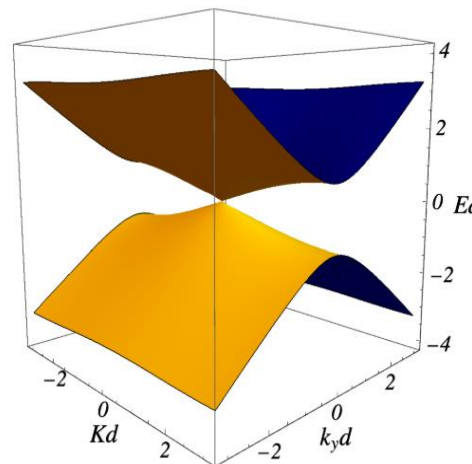
+ Mass
kink





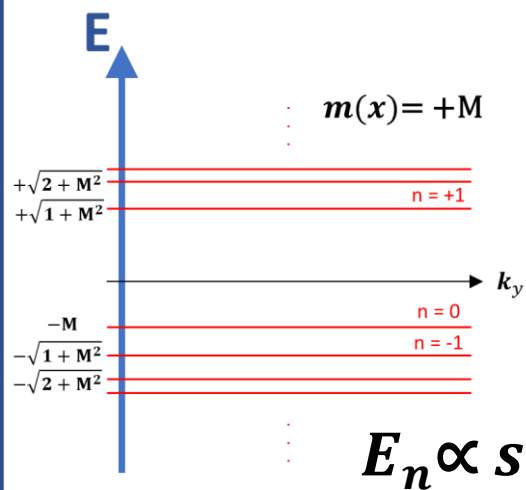
Dirac cones

+ Mass
→
super-lattice



Highly
anisotropic
Dirac cone

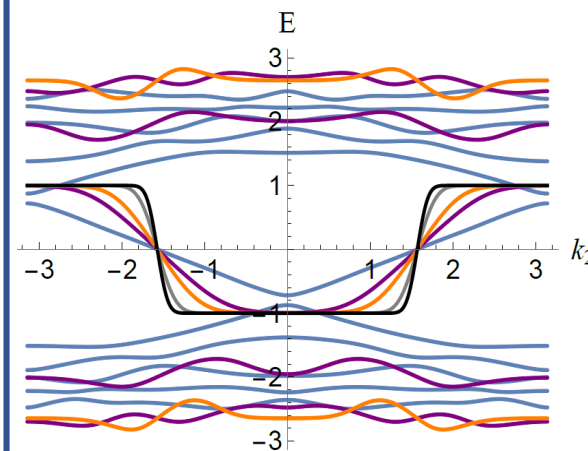
↓ +Magnetic field
+uniform mass



Degenerate
Landau levels

$$E_n \propto \text{sgn}(n) \sqrt{|n| + M^2}$$

+ Mass
→
super-lattice



Projection
on magnetic
Bloch states

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